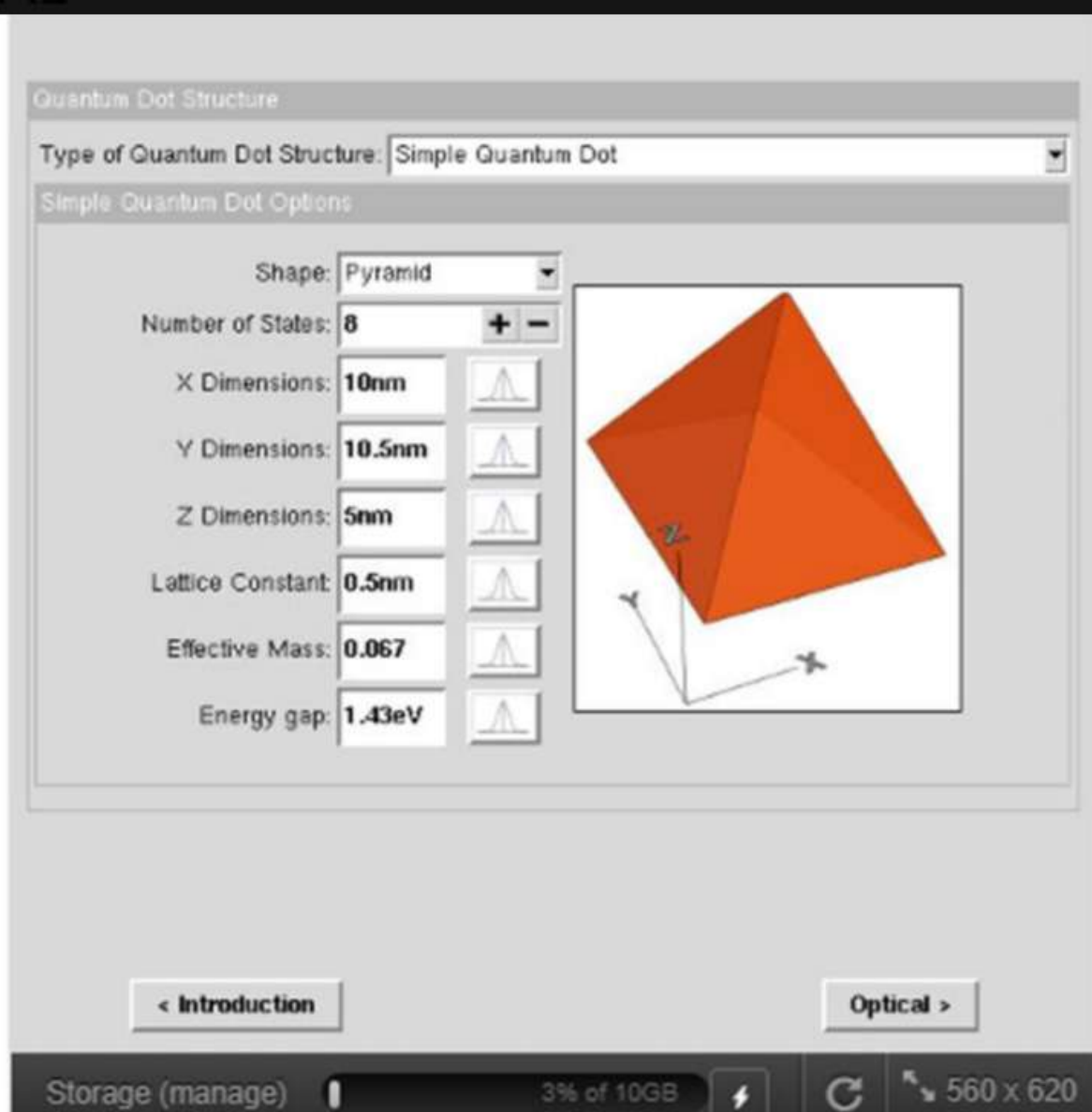


Exp. 3. Simulation of Energy States of Quantum Dot using NanoHub

T.Poojitha(192412318)

Slot-c ECA3102 NANO ELECTRONICS

OUTPUT



Light Incident Angles

Angle Theta: **45deg**

Angle Phi: **0deg**

Sweep

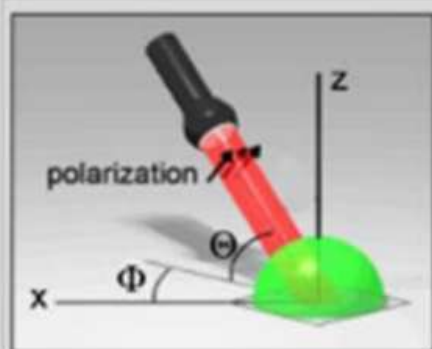
Sweep Parameter: Angle phi (deg) 

Minimum: **0**

Maximum: **90**

Number of Points: **3**  

Light Source



Absorption

Simple Q Dot - Absolute Fermi Level:    **no**

Simple Q Dot - Rel. or Abs. Fermi Level: **0eV**

Multi-Layer - Fermi Level: **0.7eV**

Temperature:  **300K**

State Broadening: **0.001**

[< Structure](#)

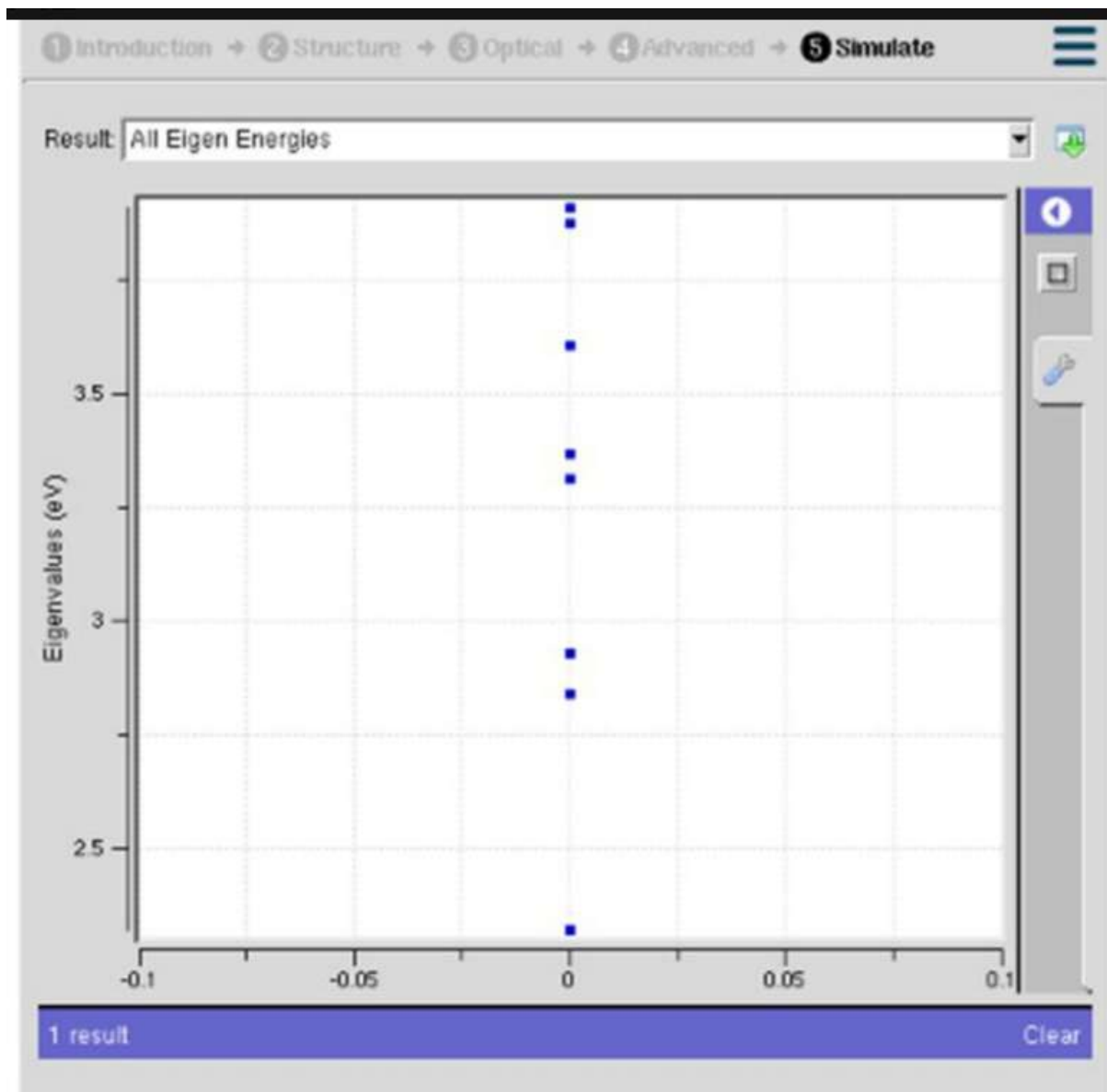
[Advanced >](#)

Storage (manage)

3% of 10GB

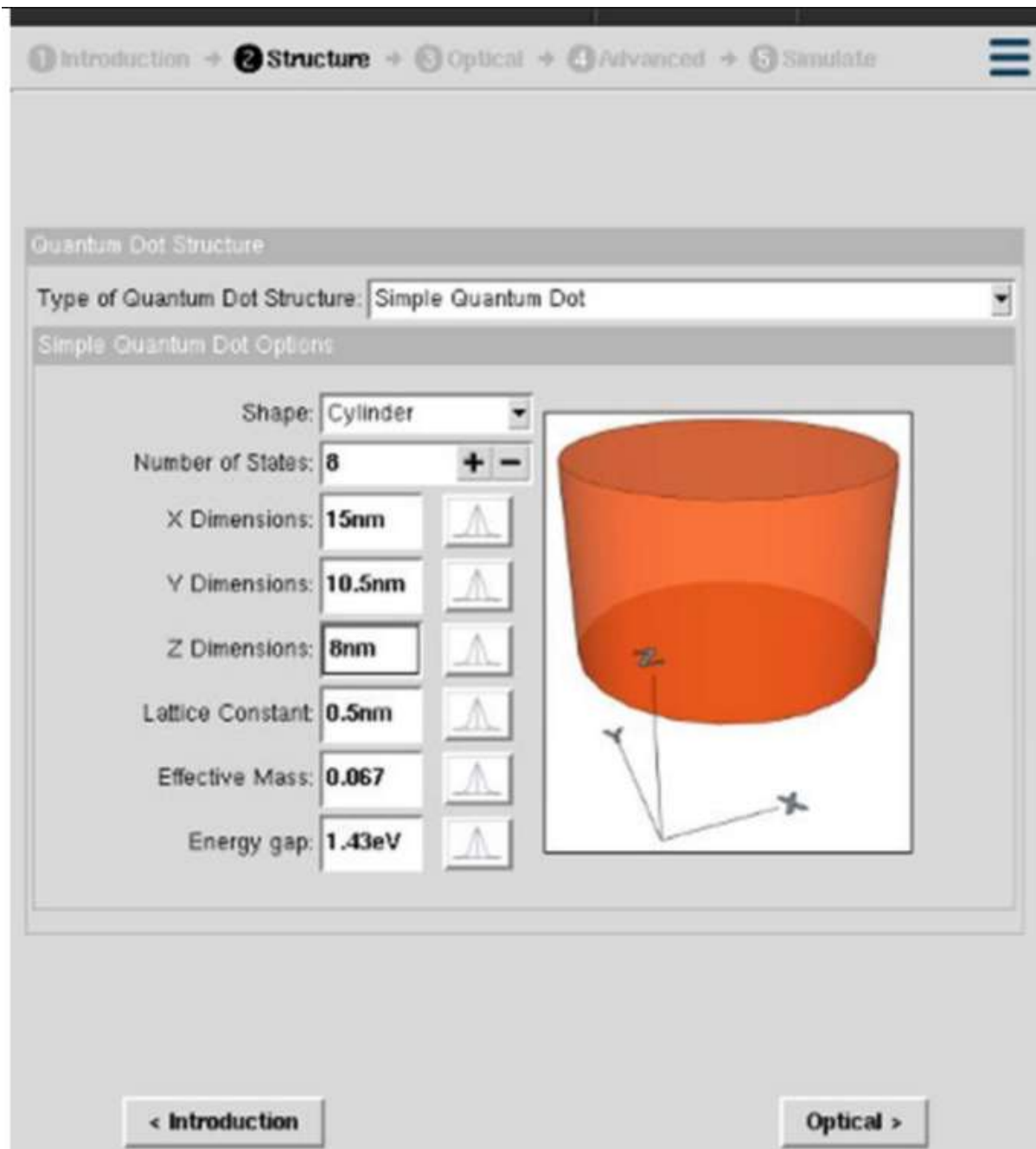


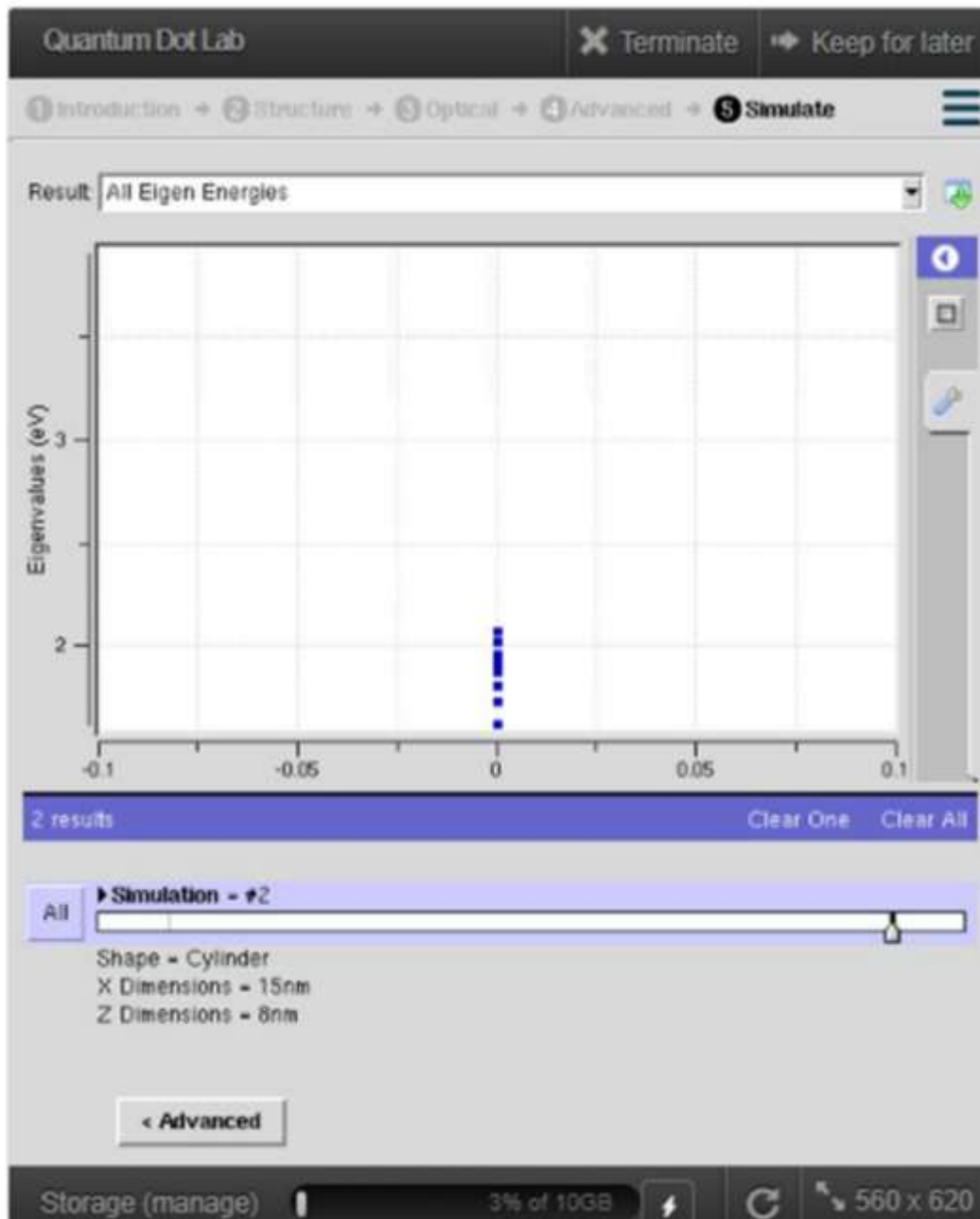
560 x 620



These figures show the complete quantum dot simulation process. First, a pyramid-shaped quantum dot is designed by defining its dimensions, material properties, and energy gap. Next, the optical simulation parameters, such as light incidence angle, temperature, and absorption settings, are configured. Finally, the simulation generates the eigen energy levels, showing the discrete energy states of the quantum dot caused by quantum confinement.

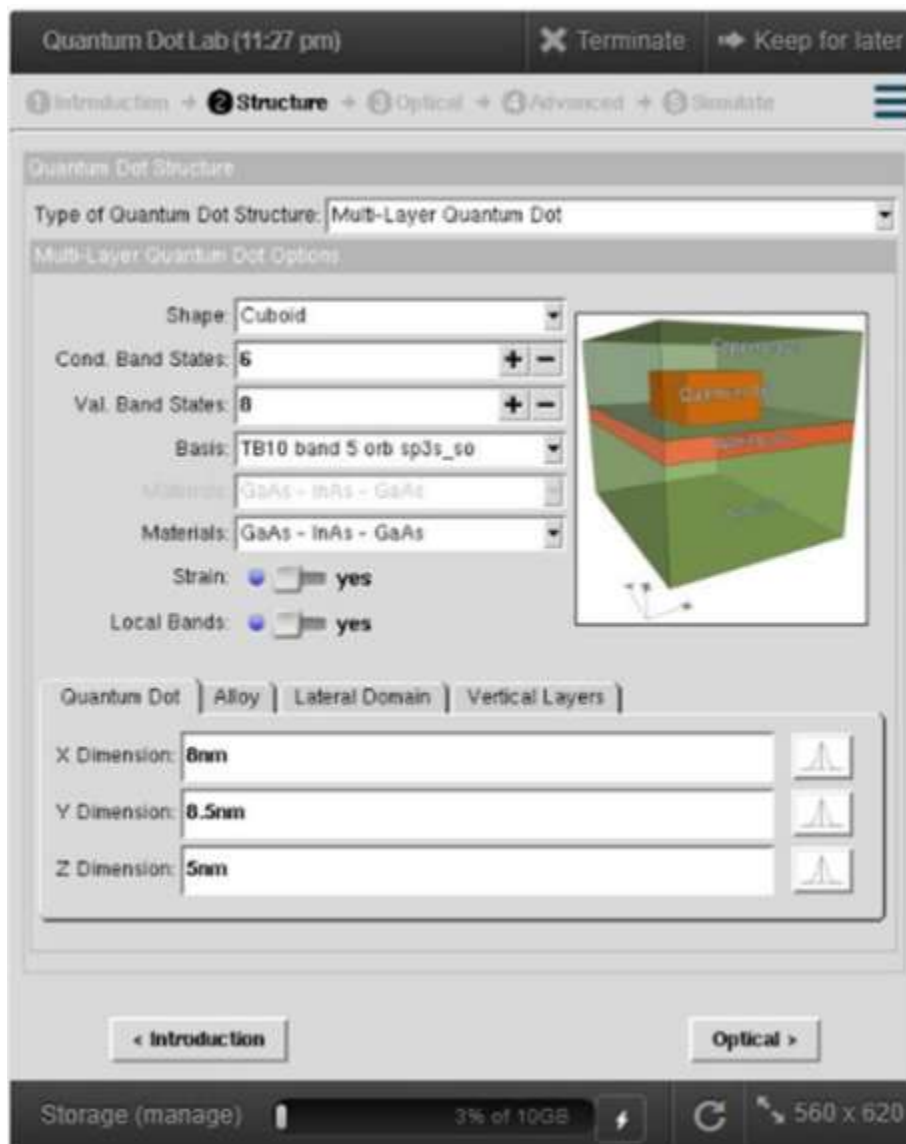
Case Study 1: Simulate the energy states of cylindrical quantum dots of $X=15\text{ nm}$, $Y = 10.5\text{nm}$ and $Z = 8\text{ nm}$.

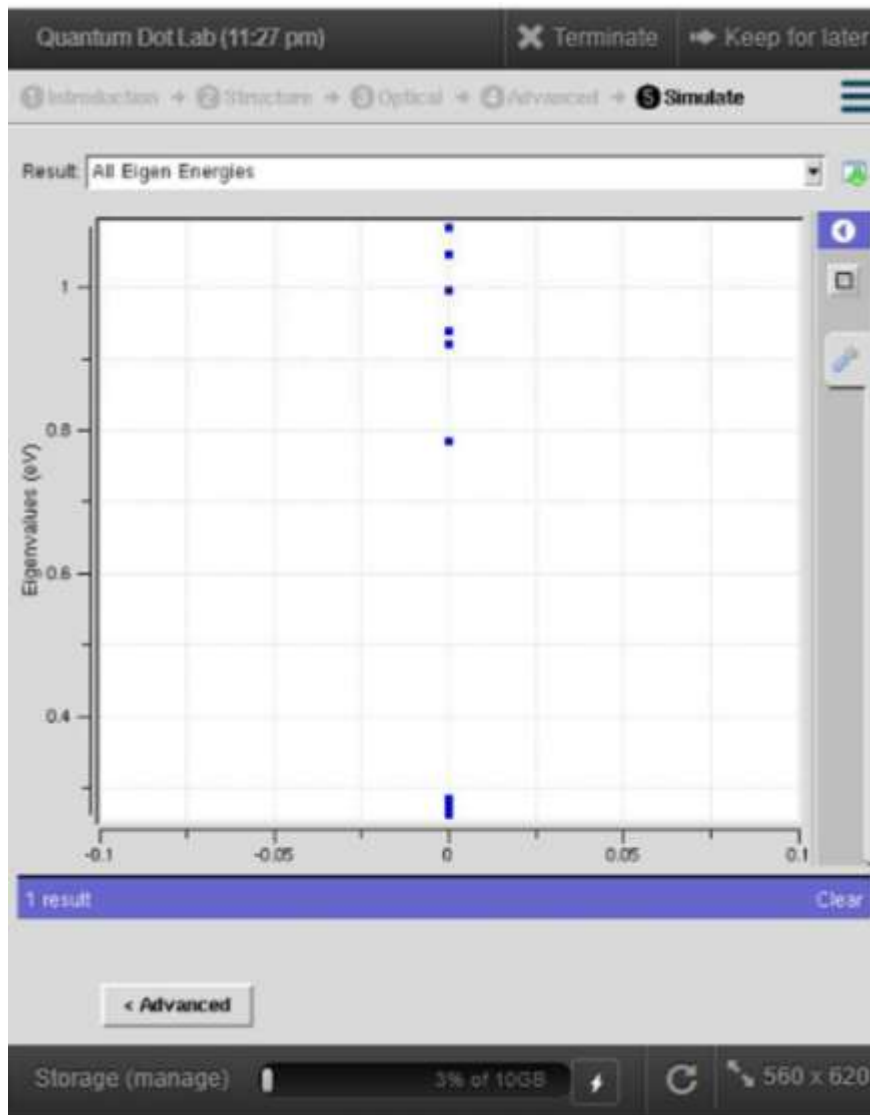




These figures show the simulation of a cylindrical quantum dot. The first image illustrates the quantum dot structure with its dimensions, material properties, and energy gap configured. The second image presents the eigen energy levels obtained from the simulation, showing the discrete energy states formed due to quantum confinement in the cylindrical quantum dot.

Case Study 2: Simulate the energy states of two layer cuboid quantum dots





These figures show the simulation of a multi-layer cuboid quantum dot. The first image illustrates the quantum dot structure with multiple material layers, dimensions, and material properties configured. The second image presents the eigen energy levels obtained from the simulation, showing the discrete energy states formed due to quantum confinement in the multi-layer quantum dot.